

Project Design Phase

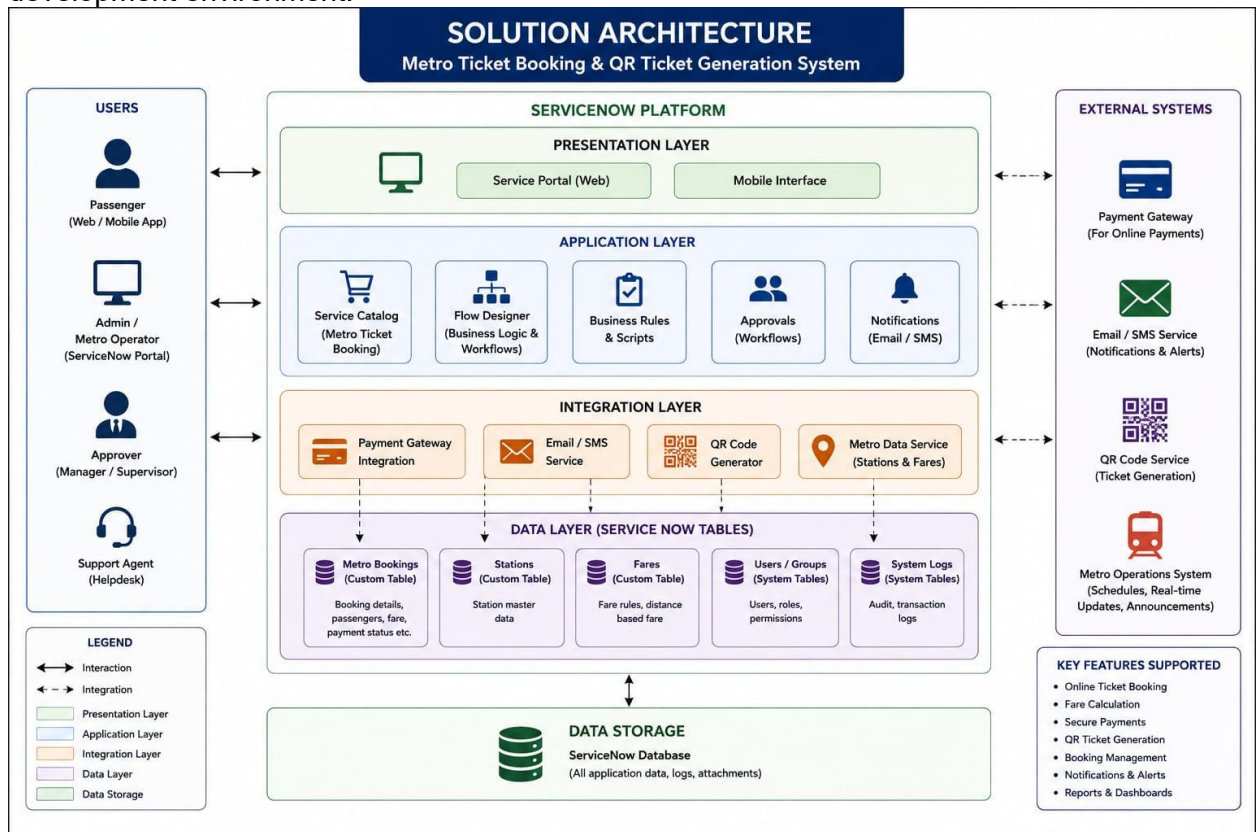
Solution Architecture

Date	27 June 2026
Team ID	
Project Name	Metro Ticket Generation System Using ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between the business problem (manual metro ticketing) and the technology solution (ServiceNow automation). The Metro Ticket Generation System architecture achieves the following goals:

- Find the best tech solution: ServiceNow platform provides all required capabilities — catalog, workflow, database, and admin — in one integrated environment.
- Describe system structure: Layered architecture from presentation through data storage within ServiceNow's cloud platform.
- Define features and phases: Service Catalog booking → validation → fare calculation → workflow automation → record storage → admin management.
- Provide specifications: All components are configured within ServiceNow's declarative development environment.



Solution Architecture Layers:

Layer	Component	Technology / Tool
Presentation Layer	Metro Ticket Booking form accessible to commuters via web browser	ServiceNow Service Portal, Service Catalog
Application Logic Layer	Input validation, fare calculation, field visibility control	Client Scripts, UI Policies
Workflow Layer	Captures form variables and automates booking record creation	Flow Designer
Backend Logic Layer	Server-side data processing, status updates, data consistency	Business Rules
Data Layer	Stores all metro booking records (route, fare, passenger, status)	ServiceNow Custom Table (Metro Booking Table)
Administration Layer	View, search, filter, and manage all booking records	ServiceNow Admin Dashboard / List View
Future Integration Layer	Payment gateway, QR code, SMS/email notifications, mobile app	REST API integrations (planned)

Architecture Data Flow:

Commuter (Web Browser) → ServiceNow Service Portal → Metro Ticket Booking Catalog Item → [Client Scripts validate & calculate fare] → [UI Policies control form fields] → Form Submission → Flow Designer triggers → Custom Metro Booking Table (record created) → [Business Rules execute backend logic] → Admin Dashboard (record visible) → Future: Payment API / QR Code / Notification Service

(Insert ServiceNow solution architecture diagram here — showing the Service Portal, Service Catalog, Client Scripts, Flow Designer, Metro Booking Table, and Admin Dashboard components with data flow arrows.)